

Deep mind-wandering amplifies attentional capture

Shibu Shelat

Attention Lab / META Lab

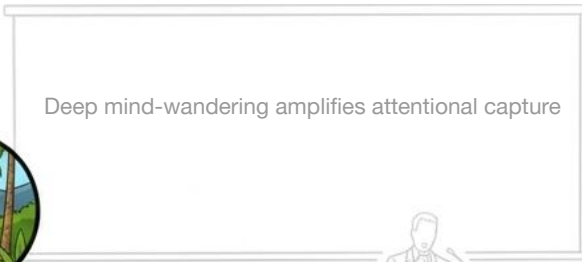
University of California, Santa Barbara

Deep mind-wandering amplifies attentional capture



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mind-wandering



Deep mind-wandering amplifies attentional capture

visual distractor



Research question:

How does mind-wandering modulate spatial attention?



Procedure

PARTICIPANTS

68 online subjects (54 female)

$M_{\text{age}} = 18.91 \pm 0.97$

all normal / corrected vision



SPATIAL ATTENTION TASK

additional singleton paradigm
with interleaved mind-wandering
depth probes



SURVEYS

to check for self-report bias in
probe responses

Procedure

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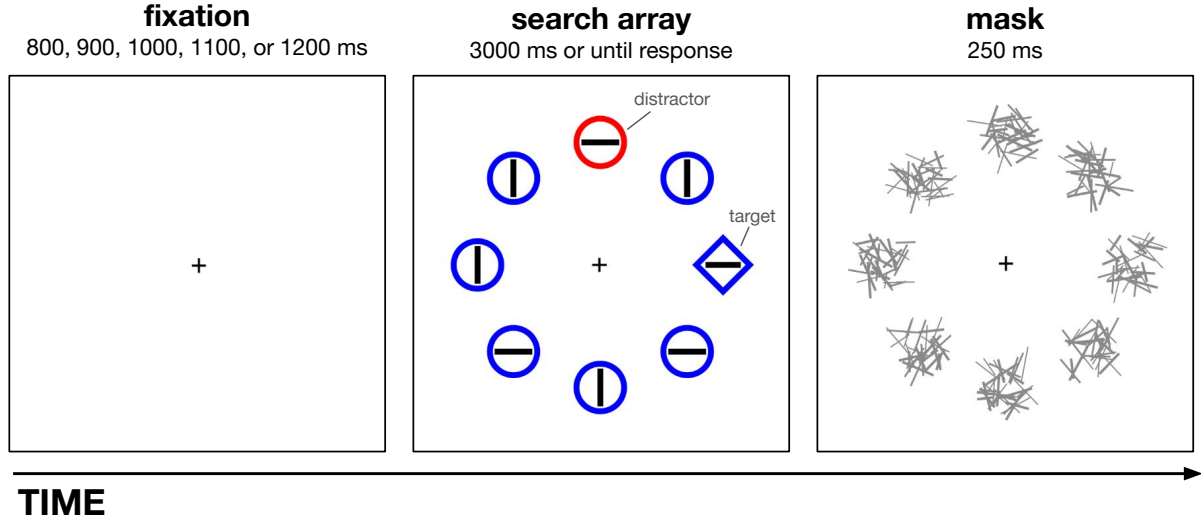
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4 blocks of 128 trials each

→ **512 total trials**

15-second break between blocks

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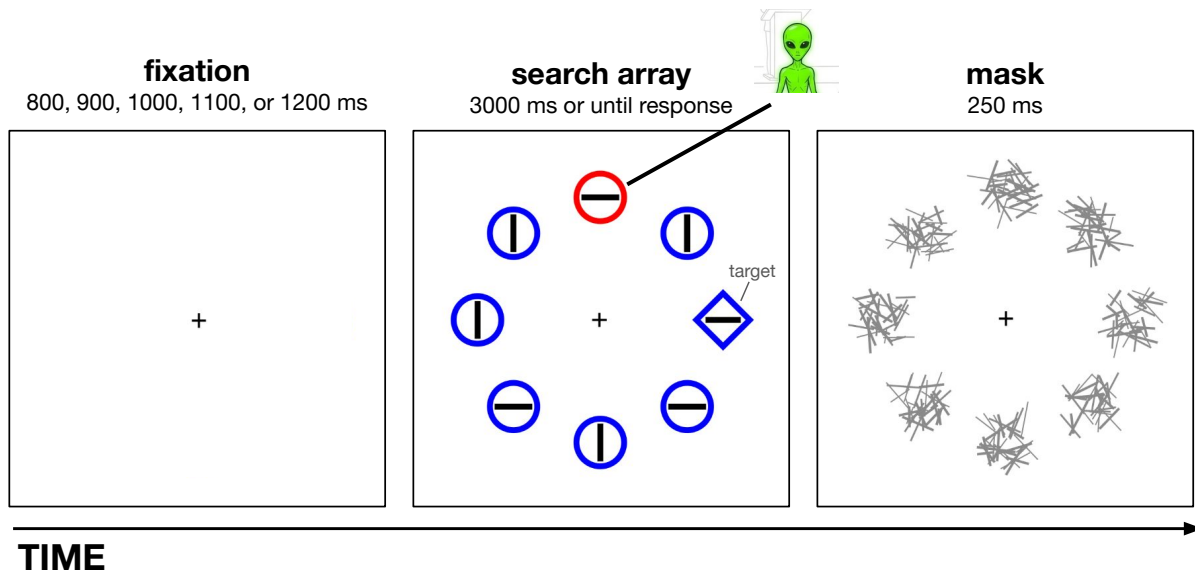
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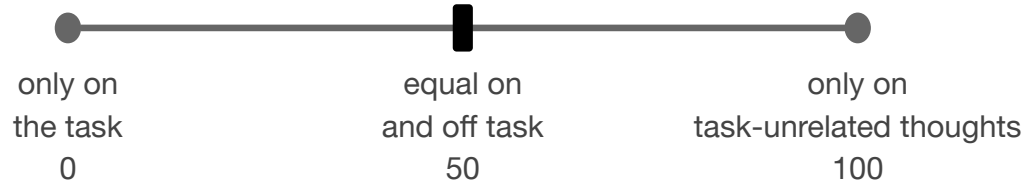
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SURVEYS

to check for self-report bias in
probe responses

*Just before the shapes appeared on the last trial,
where was your attention?*



1 probe per 16 trials
(after trial 10, 12, 14, or 16)
→ **32 total probes**

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SURVEYS

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Socially Desirable Responding on Self-Reports

Delroy L. Paulhus
University of British Columbia, Vancouver,
BC, Canada

Social Desirability Bias: A Demonstration and Technique for Its Reduction

Randall A. Gordon
Western Carolina University

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Preprint

Social desirability bias obscures mind-wandering reports

February 2026

DOI: [10.31234/osf.io/d234m_v3](https://doi.org/10.31234/osf.io/d234m_v3)

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✓ no social desirability biases
in our mind-wandering reports
 $p_s > .29$

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Statistical analysis

Logistic / linear mixed-effects regression

AIC model comparison for best-fitting random effects structure
(Burham & Anderson, 2004)

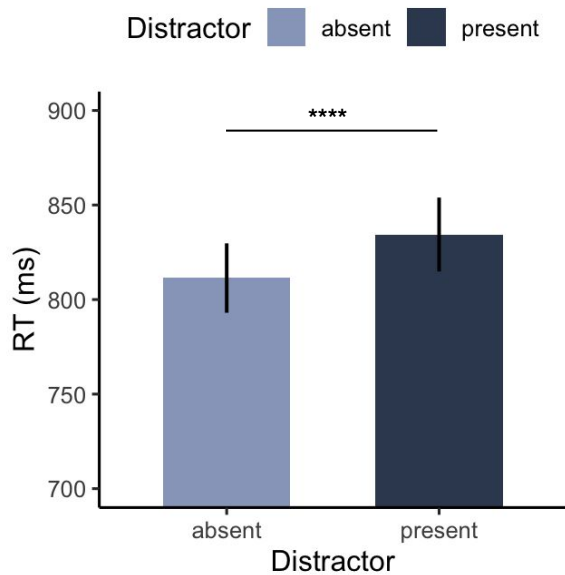
**** $p < .0001$

*** $p < .001$

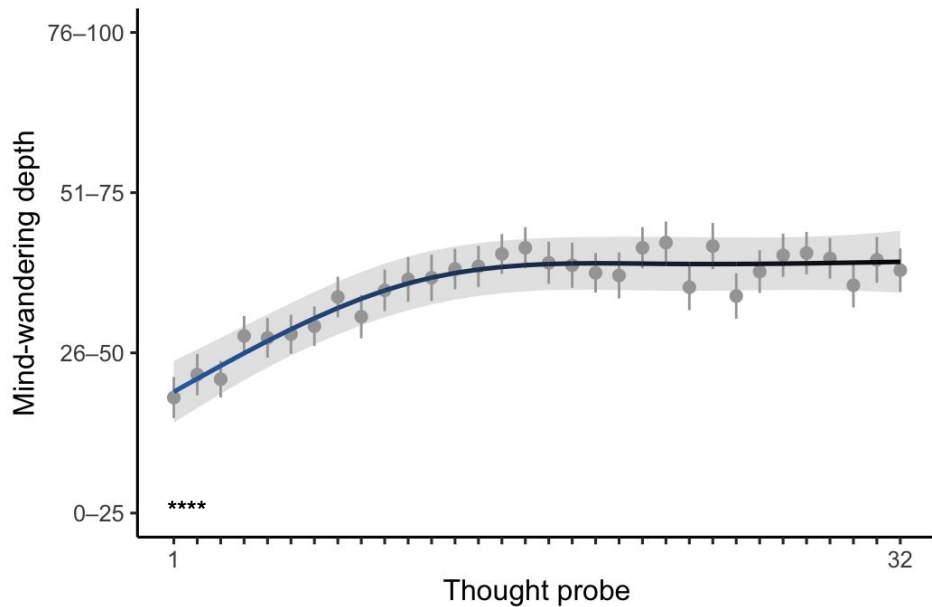
** $p < .01$

* $p < .05$

Validation checks



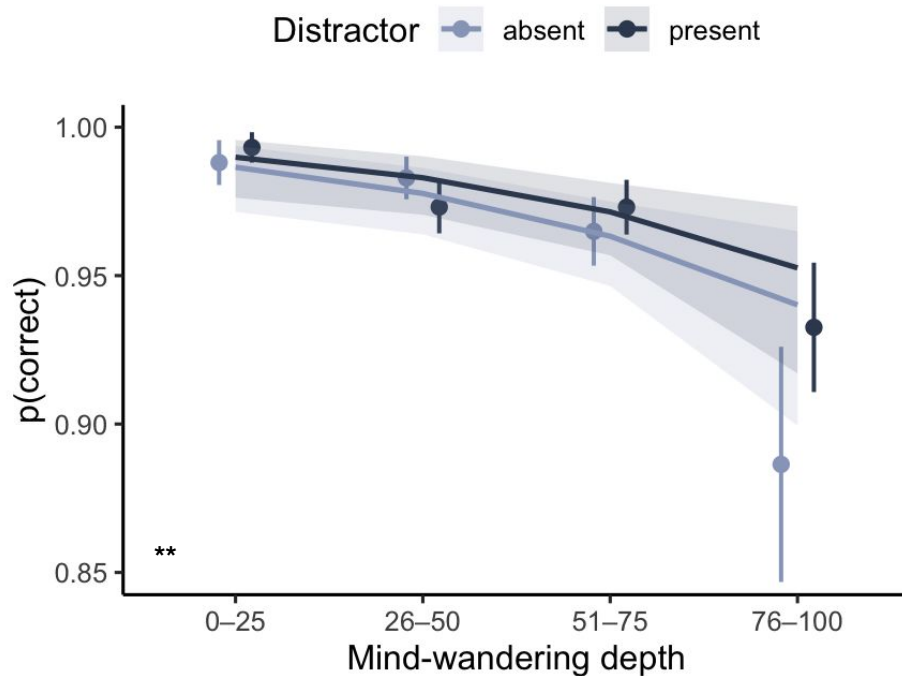
Distractor slows RT



Mind-wandering increases over time

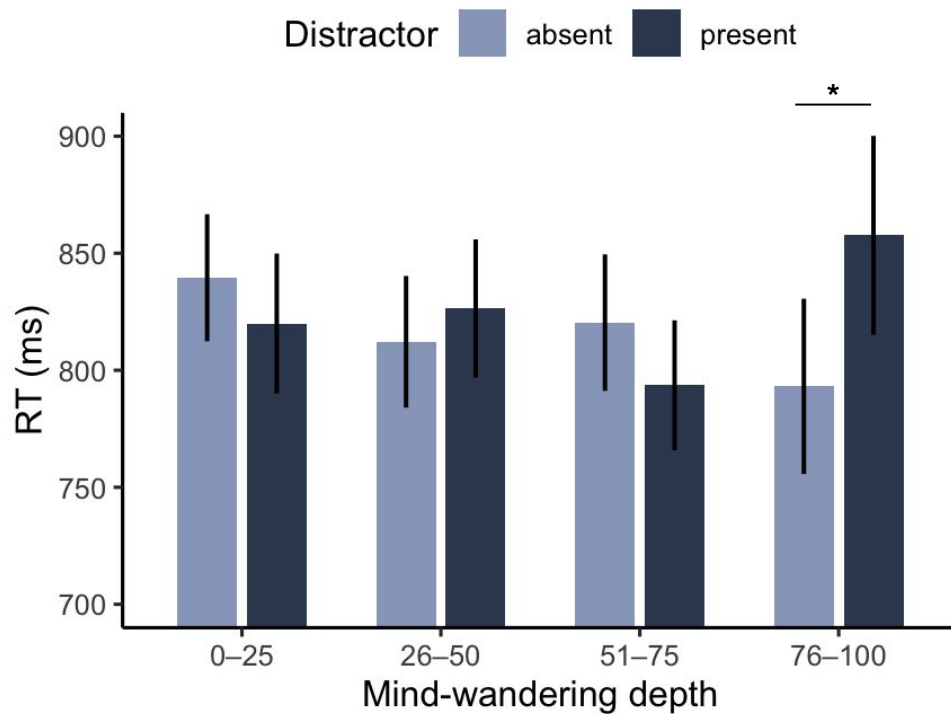
Mind-wandering → reduced accuracy

probed trials only



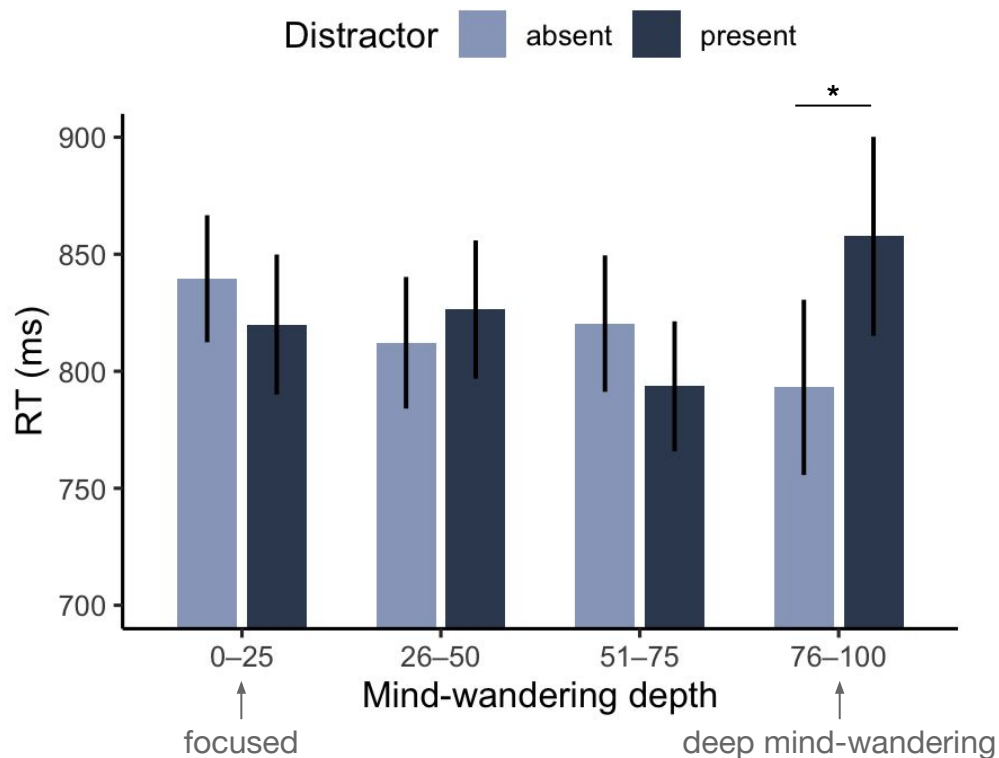
Deep mind-wandering → amplified distractor costs

accurate probed trials only

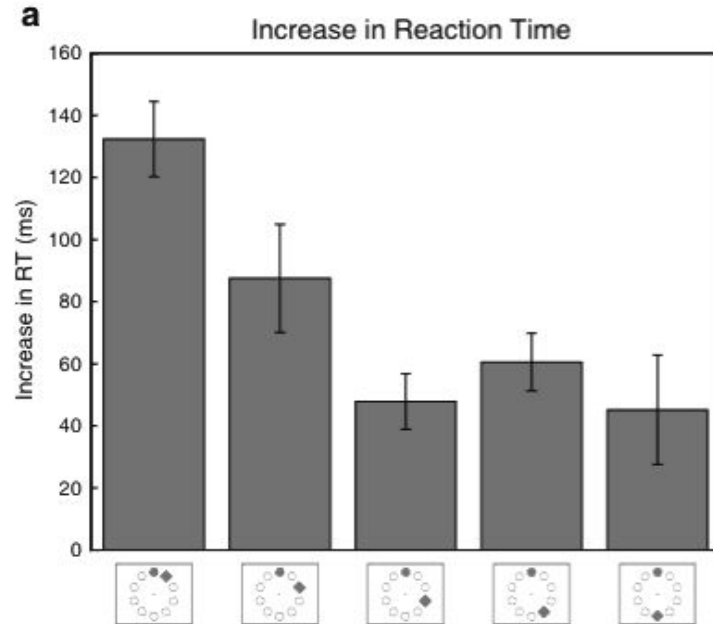


Deep mind-wandering → amplified distractor costs

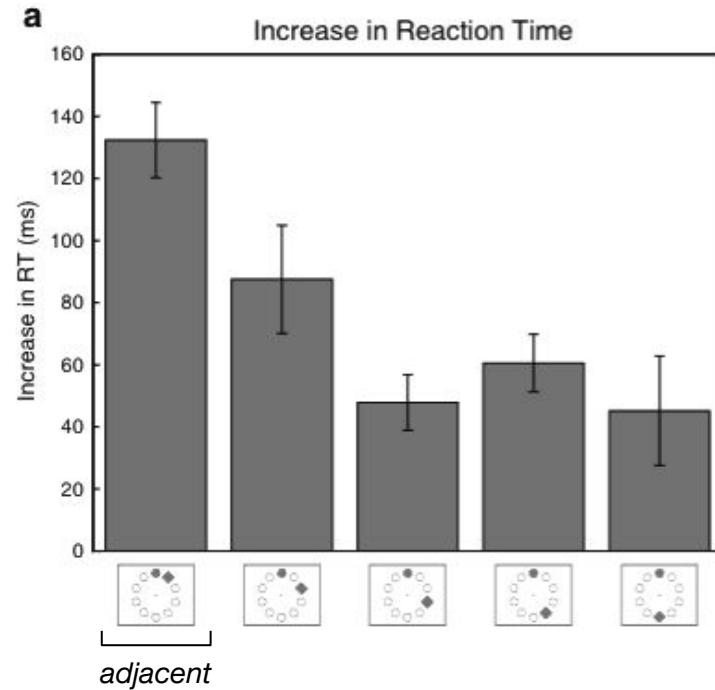
accurate probed trials only



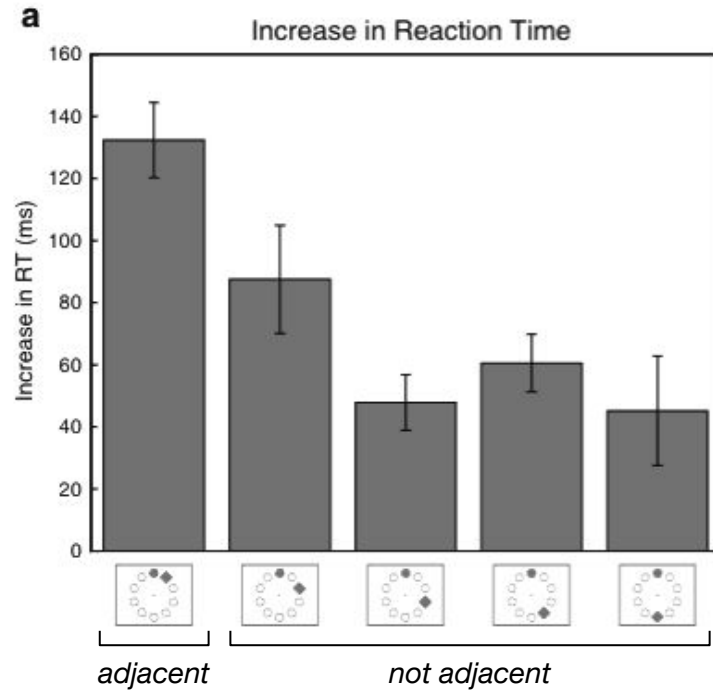
Distractor effects depend on distance from the target



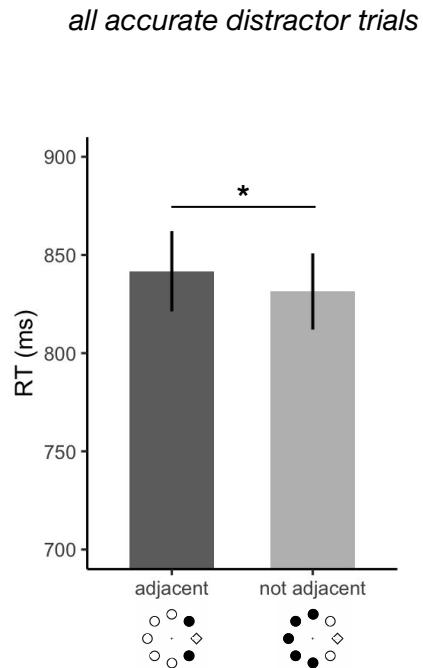
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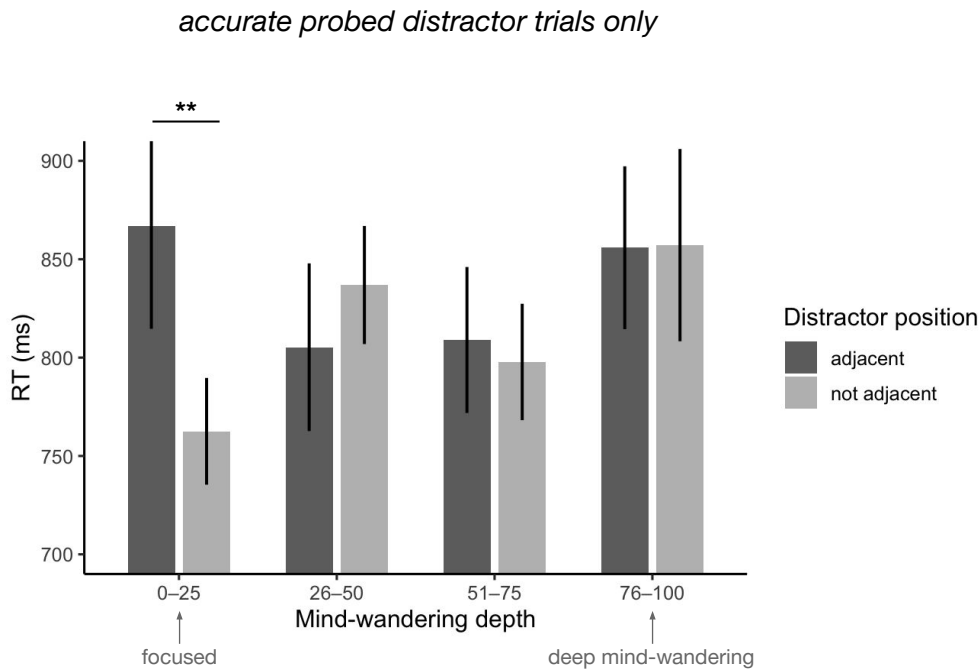
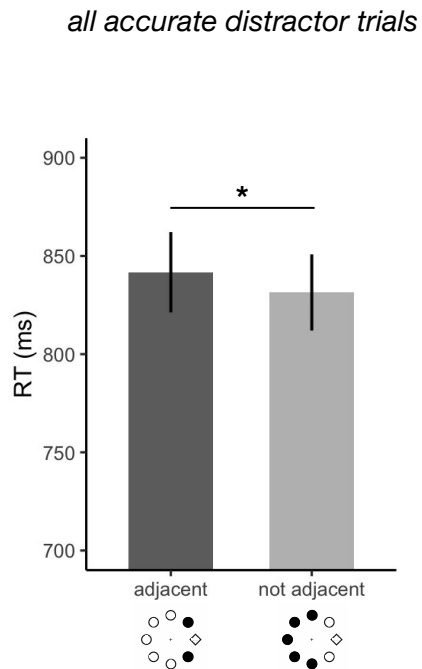
Distractor effects depend on distance from the target



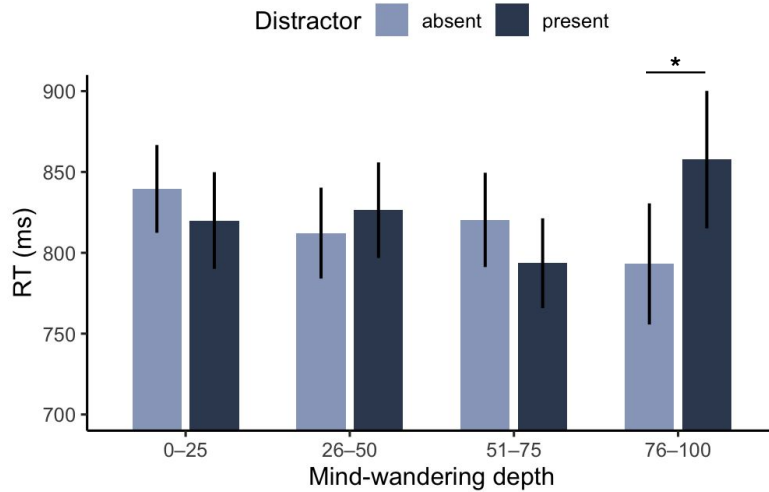
Mind-wandering → less distance-dependent distractor effects



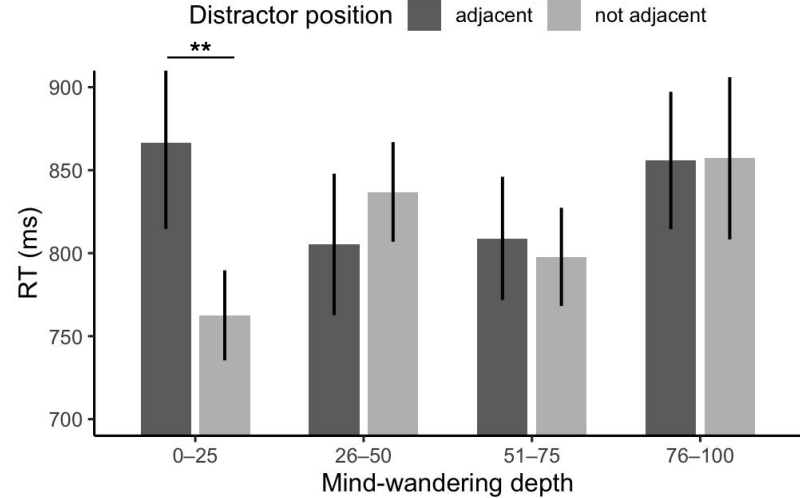
Mind-wandering → less distance-dependent distractor effects



Summary



On all trials, mind-wandering →
larger global distractor costs



On only distractor trials, mind-wandering →
less distance-dependent distractor costs

∴ Mind-wandering → diffuse spatial selectivity

Mind-wandering modulates spatial attention

During focused states:

Attention is spatially selective around the target

Distractor interference depends on distance from the target

Nearby distractors interfere more than distant distractors

Interference becomes spatially specific

During mind-wandering:

Attention is less spatially selective and more diffuse

Distractor interference becomes less dependent on distance

Nearby *and* distant distractors interfere similarly

Interference becomes global

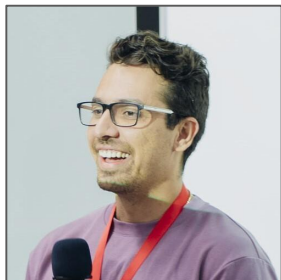
Thanks for your **attention!**



Barry Giesbrecht



Jonathan Schooler



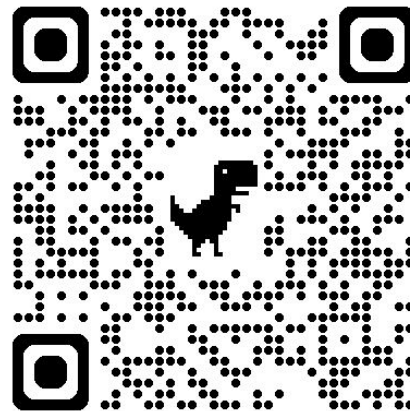
Daniel Thayer



Alien

META LAB
MEMORY EMOTION
THOUGHT AWARENESS

 **attention lab**
uc santa barbara



contact: sshelat@ucsb.edu

funding: NSF GRF 2139319

Data cleaning

removed subjects whose accuracy $< 75\%$

trimmed missing responses

trimmed trials < 200 ms

trimmed trials where search RT exceeds 3 SD from subject's mean

trimmed trials where thought probe RT exceeds 3 SD from subject's mean